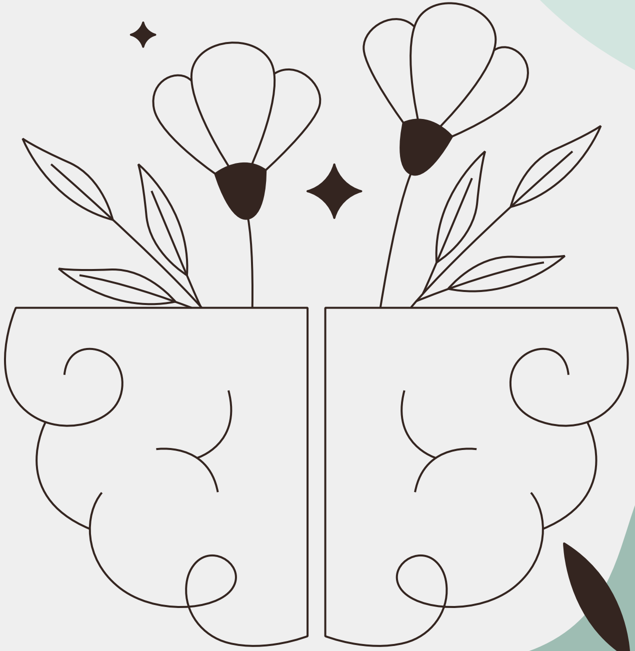
Predicting Depression Score

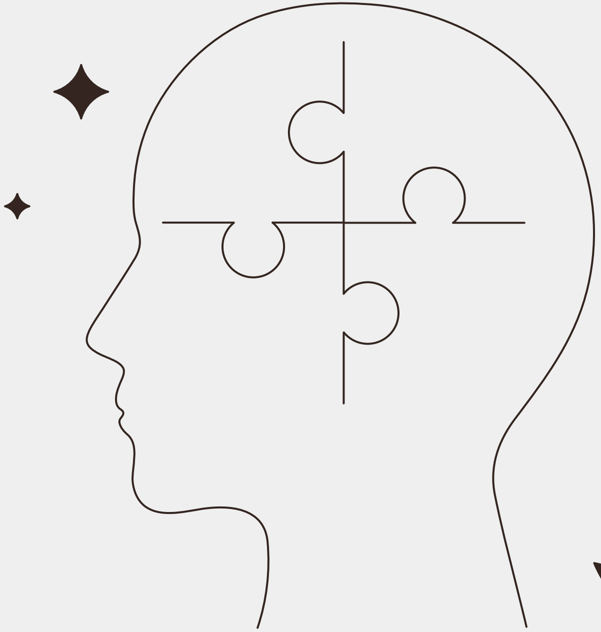
Based on Demographics, Income, Weight, Occupation, and Sleep

By Aiden Brown and Tony Nguyen



Why This Topic?

Background:
From Aug, 2021 - Aug, 2023, the CDC reported that 13.1% of U.S. adolescents and adults over the age of 12 struggle with depression, the highest prevalence was among teenagers and adolescents.



We wanted to determine if and how **factors** such as:

- income
- sleep
- demographics
- occupation

correlate to depression diagnoses.

Research Question

What impact, if any, do different demographics, sleep behaviors, incomes, weights, and occupations have on predicting depression score?

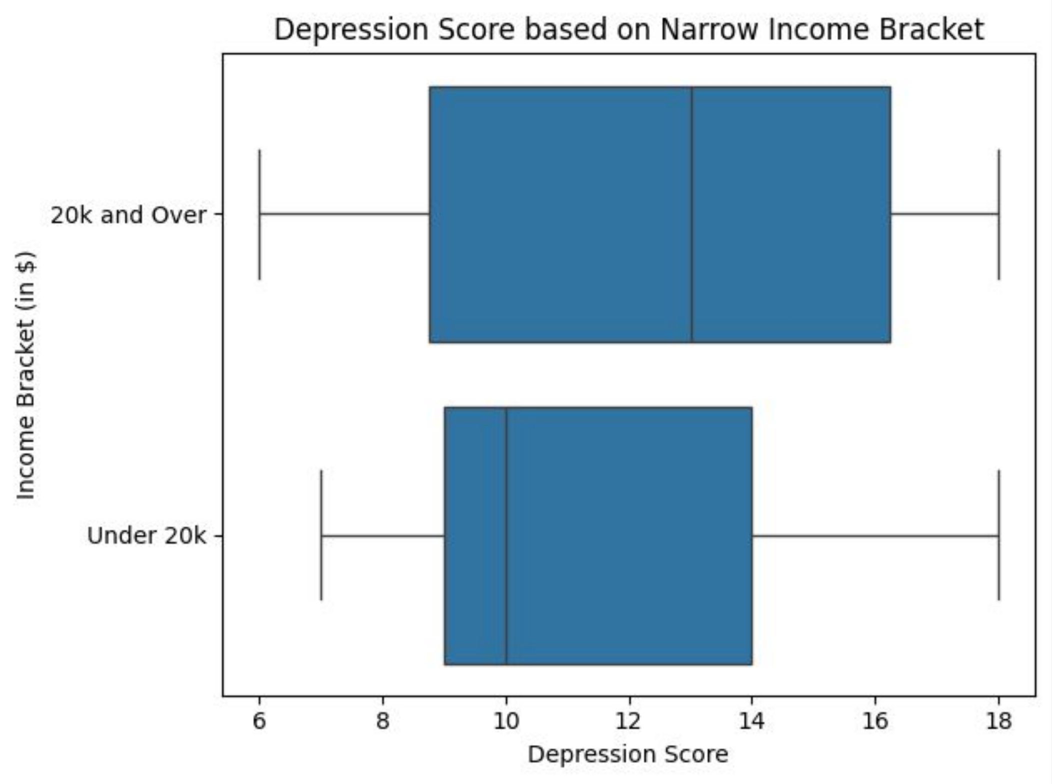
Data Source - CDC NHANES

- The CDC has annual reports on several different health conditions
- The 2017-2018 data was most accessible
- NHANES - National Health and Nutrition Examination Survey
- The CDC is one of the primary authorities on health in the US
- Depression score is an aggregation of the severity of depression symptoms

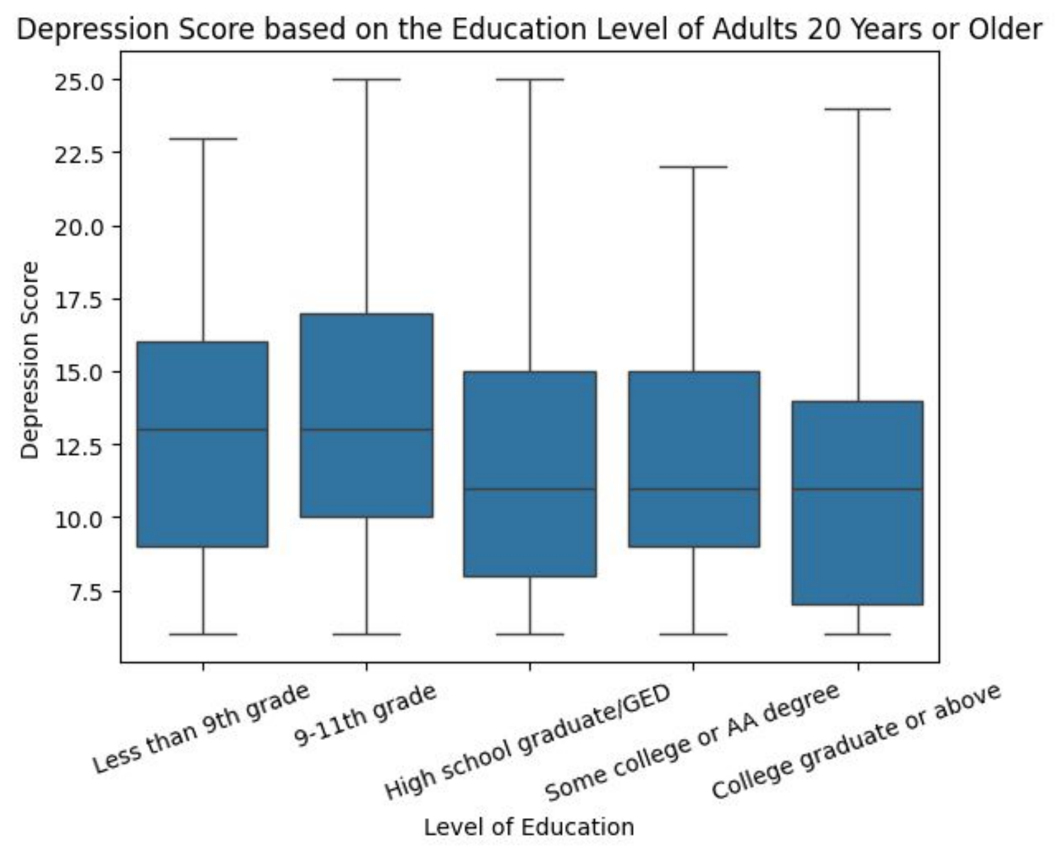
Data Cleaning and Wrangling Methods

- Cleaning:
  - Replaced invalid values for depression indicators with NaN
  - Converting variables to numeric due to file conversion
  - Replaced encoded values with NaN
- Wrangling
  - Merging 4 datasets by sequence number
  - Dropped observations with less than 6 out of 9 depression indicator questions not answered
  - Dropped observations that did not have a valid depression score

Insight from EDA: Unexpected Relationship Between Income and Depression Score



Insight from EDA: Relationship Between Level of Education and Depression Score



Predictive Models

Linear Regression & LASSO

Setup & Features:

- Trained on 5 predictors: Race, Gender, Poverty Ratio, and Weekday Sleep, and Weekend Sleep.
- LASSO utilized 10-fold cross-validation.
- Applied standardization to all variables.

Neural Network

Architecture:

- Trained for 50 epochs using the same 5-feature subset.
- Two linear layers with ReLU activation.
- Implemented a 40% dropout rate for regularization.

Gradient Boosting

Dual-Model Approach:

- Regressor for continuous depression scores.
- Classifier for binary target (Score >= 10).
- XGBoost models used all features with one-hot encoding.

Results of Regression Models/Comparison

1	**Linear Regression**	RMSE: 4.39 R^2: -0.034
2	**LASSO**	RMSE: 4.39 R^2: -0.034
3	**Neural Network**	RMSE: 4.6 R^2: -0.15
4	**Gradient Boosting**	RMSE: 4.48 R^2: -0.022

Overall Interpretation:

All models yielded negative R^2 values, indicating they performed worse than predicting the mean. The low R^2 suggests that the selected predictors (Race, Gender, Poverty Ratio, Sleep) may not meaningfully capture the correlation to depression scores in this dataset.

The Gradient Boosting model showed a slight improvement in R^2 compared to other models, suggesting a potential for other variables to be explored further.

Results of Regression Models/Comparison

	Model	Metrics
1	Linear Regression	MAE: 3.60 R^2: -0.034
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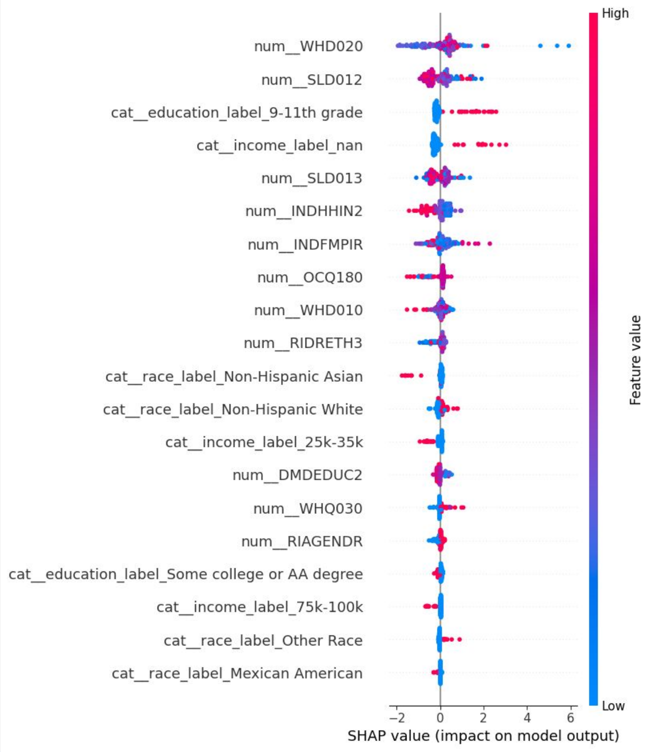
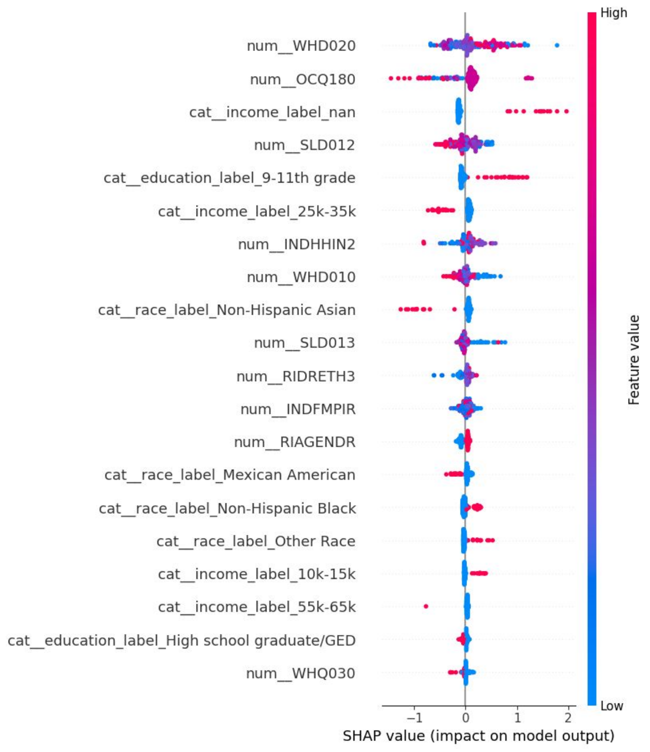
However, the Neural Network showed a slight improvement in R^2 compared to simpler linear models, suggesting a potential for non-linear relationships to be explored further.

SHAP Value Analysis

Feature Impact on Regression vs. Classification Models



Regression Model



Classification Model (Depression > 10)

Limitations

1. Only used Data from US 2017-2018

2. We selected predictor variables based on our hypotheses

3. Depression Score is not Depression Diagnosis

4. Data we (necessarily) dropped may have changed relationships

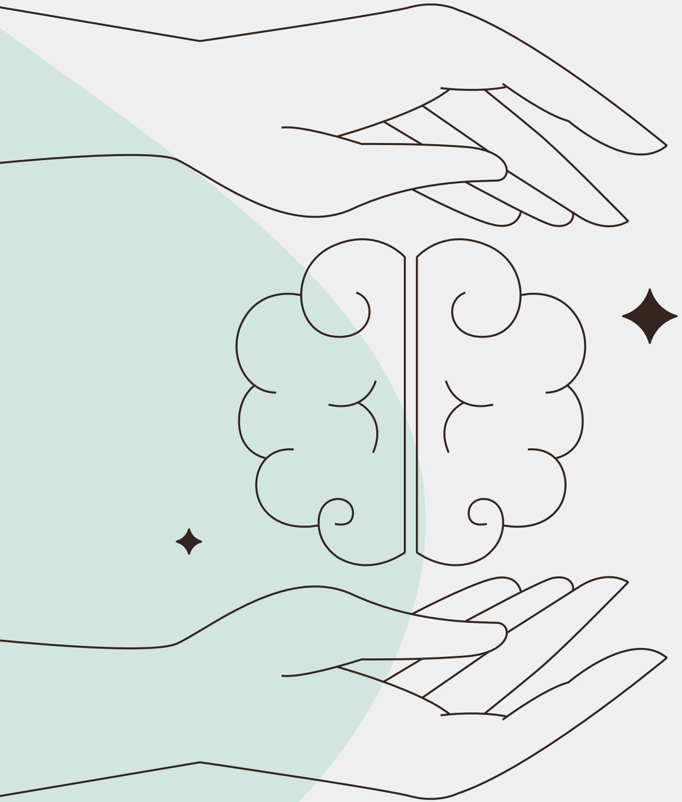
Conclusions and Future Work

Conclusions

- EDA suggested possible correlations between predictors and depression score but our models had weak performance
- Our predictor variables were not strong enough on their own to predict depression score

Future Work

- How will our models perform with data from other years and other nations?
- Increase quantity and variety of predictor variables
- Compare depression score and another metric of depression? Are the same predictors relevant?



Thanks!

What questions do you have?

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